

Assignment - javascript

Course ID: 69500952eeb7c3979890f9c8

Generated: 12/27/2025

Due Date: 1/3/2026

Questions: 5

Type: DESCRIPTIVE

ASSIGNMENT QUESTIONS:

Javascript Programming Fundamentals Assignment

Instructions: Please answer the following questions thoroughly and provide code examples where requested to support your explanations.

1. Explain the key differences between `var`, `let`, and `const` in JavaScript, focusing on their scope (global, function, block) and reassignability/re-declaration. Provide a concise code example for each to illustrate your points.
2. Describe what a callback function is in JavaScript and explain its primary use case in asynchronous programming. Provide a simple code example demonstrating how a callback function can be used, for instance, with `setTimeout`.
3. How can you dynamically manipulate the Document Object Model (DOM) using JavaScript? Explain the steps to select an HTML element by its ID, change its text content, and then modify one of its CSS properties (e.g., color or font size). Provide a code snippet for both the HTML and JavaScript parts.
4. What is the purpose of Promises in JavaScript? Explain how Promises help manage asynchronous operations more effectively than traditional callback-based approaches (e.g., "callback hell"). Briefly describe the states a Promise can be in and the role of `.then()` and `.catch()` methods.
5. Describe the concept of prototypal inheritance in JavaScript. How does it work and how does it differ fundamentally from class-based inheritance models found in languages like Java or C++? Provide an example of how you would set up a prototype chain between two JavaScript "objects" (or constructor functions).

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